# Responsible Usage of AI in Scientific Research

By

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# NATIONAL GUIDELINES FOR TRUSTWORTHY AND RESPONSIBLE AI

EGYPT

Establishes a national reference for responsible development, deployment, and oversight of AI systems across public and private sectors.







## PURPOSE OF THE GUIDELINES



### WHAT SHOULD BE GOVERNED
Guide to AI Governance Framework




Risk Tiers & Mandates

### HOW TO COMPLY
The Guidelines



Methodologies, Metrics & Checklists to translate Principles into Governance realities.

### DESIGNED FOR:
Data Scientists, Compliance Officers, Developers

## GUIDELINES SEEK TO:



PROTECT INDIVIDUALS' RIGHTS & SOCIETAL WELL-BEING.



ENSURE ETHICAL & TRUSTWORTHY USE OF AI.



STRENGTHEN INSTITUTIONAL ACCOUNTABILITY & TRANSPARENCY.



PROMOTE INNOVATION GROUNDED IN SAFETY & SOUND GOVERNANCE.



ALIGN EGYPT WITH LEADING INTERNATIONAL STANDARDS.

GOVERNMENT

PUBLIC & PRIVATE SECTORS

# The Pillars of Egypt's AI Governance

Core structural and ethical requirements for trustworthy and responsible AI implementation

## INTERNAL & TECHNICAL OVERSIGHT

### 1 Institutional Leadership & Auditing

Orgenizations must appoint Chief AI Officers and maintain audit mechanisms for national regulation alignment.



### 2 The TEVV Lifecycle Protocol




Systems require Testing, Evaluation, Verification, and Validation (**TEVV**) before and after deployment.

### Embedded Trustworthiness

Responsible AI principles must be integrated starting from the initial design and development phase.

## STAKEHOLDERS & SOCIAL IMPACT

### 3 Frugal AI for Sustainability

Implementing AI that reduces computational costs and minimizes the overall environmental footprint.



### 4 Inclusive Stakeholder Collaboration

Fostering transparent communication between government, industry, academia, and civil society groups.

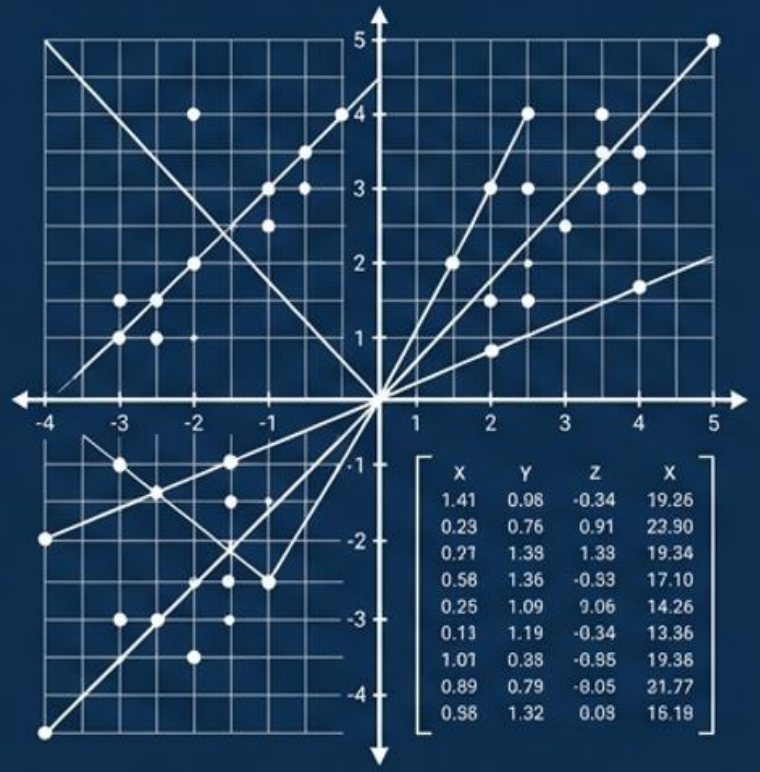


### Cultural & Social Safeguards

Aligning AI systems with cultural values while protecting vulnerable groups and societal interests.

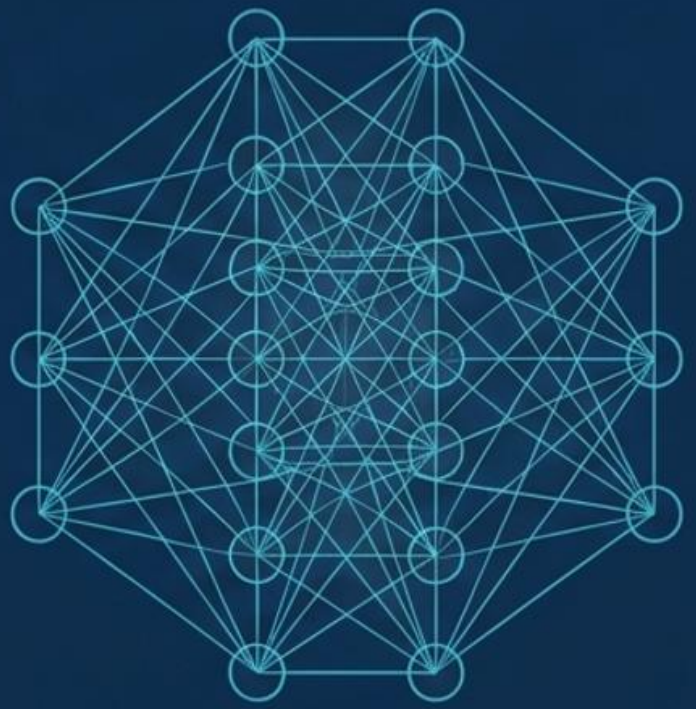


# AI in Research: Innovation Meets Academic Integrity




| X | Y | Z | X |
|---|---|---|---|
| 1.41 | 0.08 | -0.34 | 19.26 |
| 0.23 | 0.76 | 0.91 | 23.90 |
| 0.21 | 1.38 | 1.38 | 19.34 |
| 0.58 | 1.36 | -0.53 | 17.10 |
| 0.25 | 1.09 | 9.06 | 14.26 |
| 0.13 | 1.19 | -0.34 | 13.36 |
| 1.01 | 0.38 | -0.85 | 19.36 |
| 0.89 | 0.79 | -8.05 | 21.77 |
| 0.98 | 1.32 | 0.08 | 15.19 |

**Responsible Frameworks**



## THE TRANSFORMATION

AI is revolutionizing scientific discovery, data analysis, and academic writing at an unprecedented scale.

## THE HOOK

With unprecedented efficiency comes unprecedented risk to the core foundations of scientific integrity and trust.

## THE GOAL

Harness Generative AI as a powerful, assistive tool without compromising originality, accountability, or ethical standards.

# The Hidden Threats to Scientific Integrity



## Fabrication

AI models can generate plausible but entirely fake citations, datasets, and conclusions.



## Amplification of Bias

Uncritical reliance reinforces societal stereotypes, sycophantic behaviors, and biased academic consensus.



## Data Vulnerability

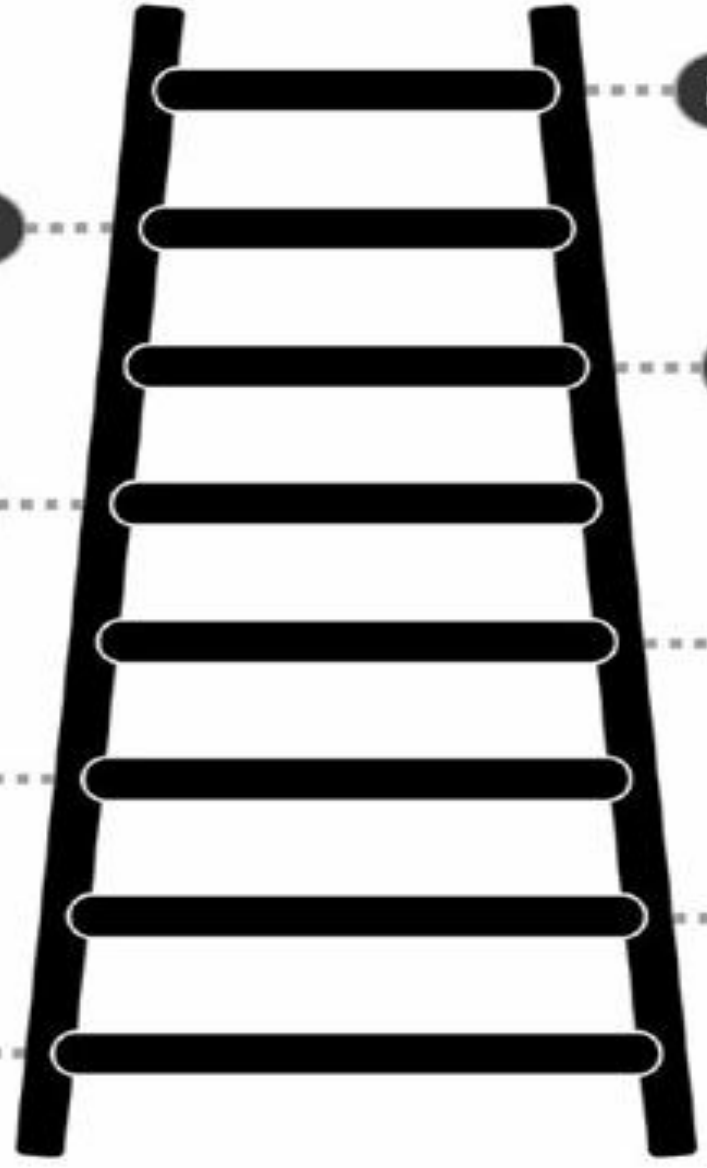
Inputting sensitive, unpublished research into open models risks immediate intellectual property loss and public disclosure.



## Erosion of Skills

Over-reliance on generative algorithms degrades critical thinking, independent reasoning, and deep analysis abilities.

# The CARE Principles for Responsible AI



**Broader Impact**
8 Awareness of social and environmental implications of GenAI use

**Transparency**
7 Transparent disclosure of GenAI use

**Accountability**
6 Claiming accountability for all GenAI based outcomes

**Bias Mitigation**
5 Preventing the perpetuation, amplification, or introduction of undesirable biases

**Originality**
4 Absence of plagiarism and adherence to scientific integrity

**Quality Control**
3 Human assessment of adherence to pre-set quality criteria

**Data Security**
2 Protection of private and sensitive data

**Regulations**
1 Adherence to international, national, institutional, and publisher regulations

# Clear Boundaries: Permitted vs. Prohibited Uses

| Permitted (Assistive Use) | Prohibited (Generative Use) |
| --- | --- |
| ✓ Correcting grammar, academic style, and language formatting. | ✗ Generating substantive scholarly content, theoretical frameworks, or analysis. |
| ✓ Summarizing original text already authored by the student. | ✗ Fabricating statistical results, research data, or academic references. |
| ✓ Suggesting structural organization and brainstorming topics. | ✗ Generating peer review reports or grant proposals completely by AI. |
| ✓ Explaining general concepts (Non-generative conceptual support). | ✗ Using AI to take exams, quizzes, or complete full assignments. |

# The Authorship Rule: Humans Remain Accountable




**AI is Not an Author:** AI systems cannot be listed as authors or co-authors under any circumstances; they lack agency and liability.

- **Final Accountability:** The human author bears full legal, ethical, and academic responsibility for accuracy and validity.
- **Mandatory Verification:** All AI outputs must be fiercely scrutinized, cross-validated, and verified against primary sources.
- **No Blind Trust:** AI systems are not liable agents and cannot be sanctioned for their errors or hallucinations.

# Mandatory Disclosure and Transparency Standards

## 1. Declare All Usage

Formal disclosure is strictly mandatory for theses, proposals, and submitted coursework.

## 2. Specify the Details

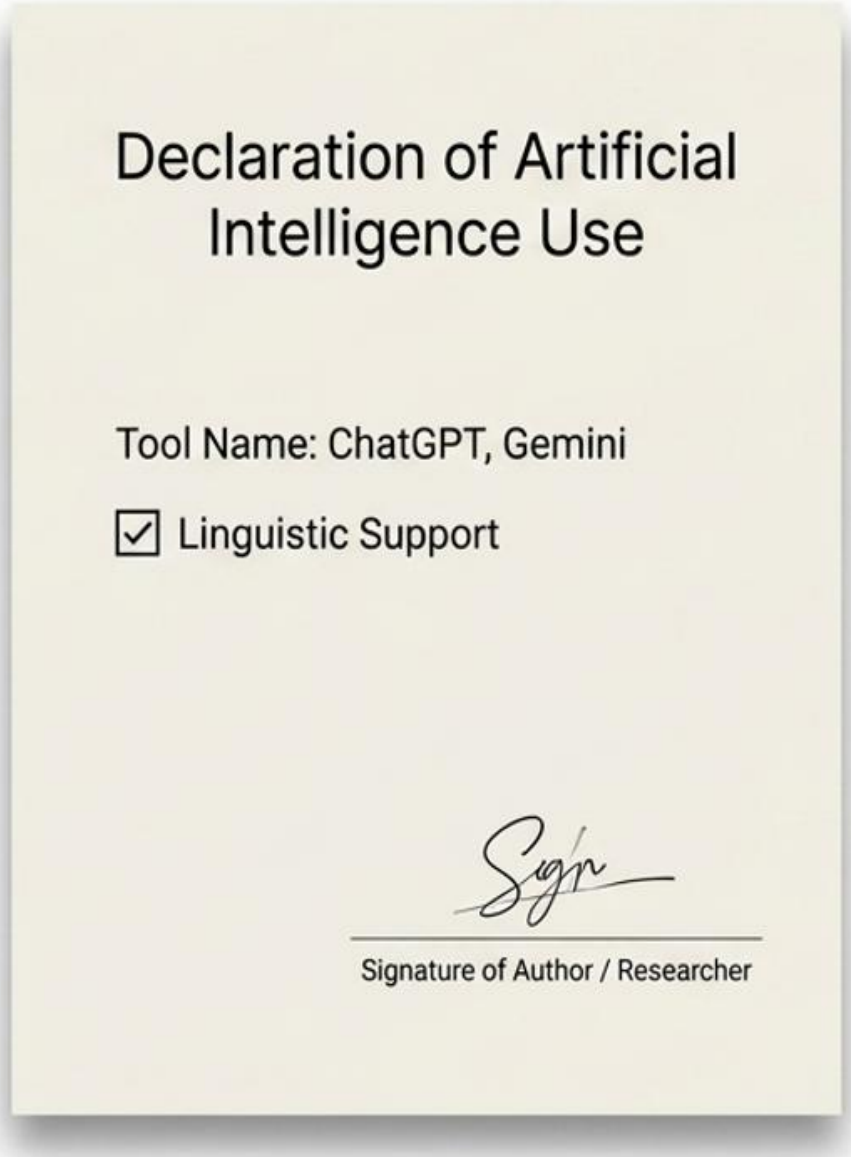
Always cite the specific tool used, version, date, and exact scope of use (e.g., 'language editing').

## 3. Location in Text

Transparently note AI usage in the methodology section or via required institutional declaration forms.

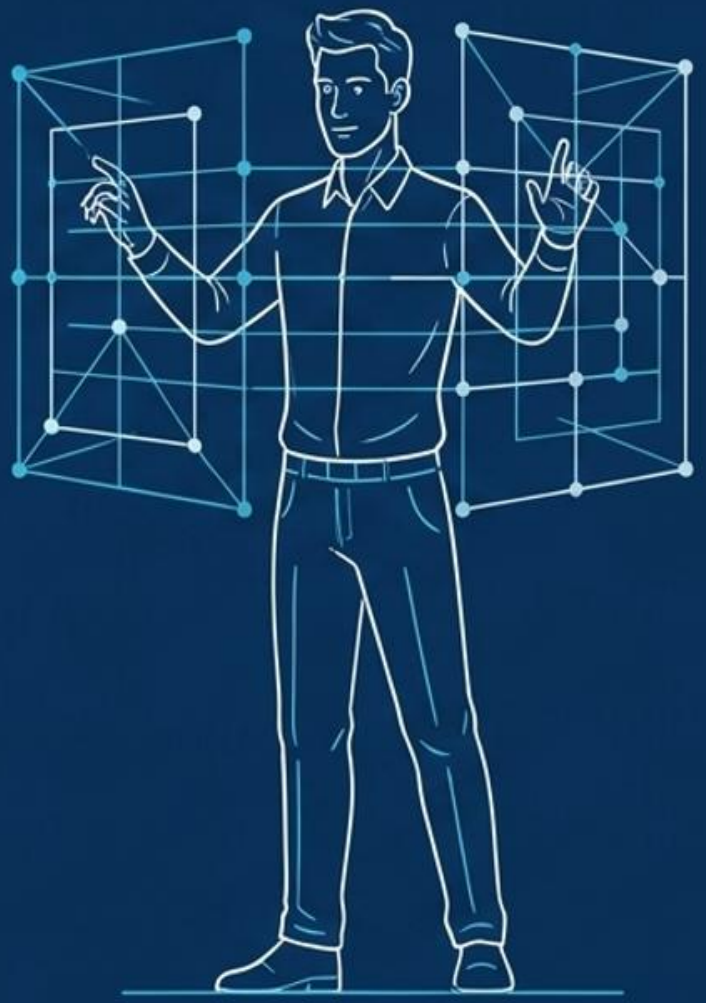
## 4. Avoid Subterfuge

Passing off AI-generated text or analysis as original human work is a severe violation of academic integrity.



# The Responsible Researcher's AI Checklist

- [ ] **Compliance:** Does my use fully align with university, publisher, and funding agency policies?
- [ ] **Privacy:** Is my input data totally stripped of sensitive, proprietary, and identifiable information?
- [ ] **Originality:** Have I independently verified every single citation, fact, and claim generated?
- [ ] **Substance:** Did I use AI strictly to assist my work, rather than to generate my core analysis?
- [ ] **Transparency:** Have I fully and accurately disclosed the specific AI tools and versions used in my methodology?



# Plagiarism in Scientific Research



# Plagiarism in Research: Beyond Copy and Paste

To educate researchers and students on the different forms of plagiarism and the underlying reasons it occurs to promote academic integrity.

Plagiarism in scientific research involves more than just stealing words; it includes the unauthorized use of ideas, poor citation habits, and lack of administrative approval. Understanding the spectrum of plagiarism—from unintentional "sprinkling" to wholesale copying—is the first step toward avoiding academic misconduct.

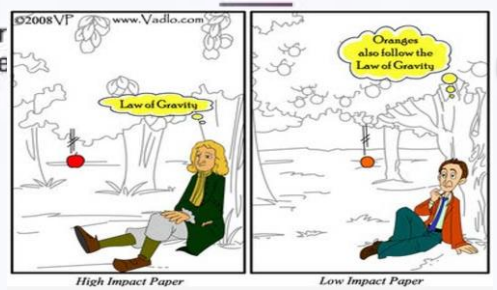
## The Spectrum of Academic Plagiarism



### Wholesale and Patchwork Plagiarism

Copying entire sources or stitching together multiple portions of work without original contribution.

### Idea-Based Plagiarism




### The "Sprinkle" Trap

Occasional failure to cite properly due to punctuation errors; often unintentional but still plagiarism.

## Root Causes and Prevention

### Common Excuses



**Primary Examples:**
Boring assignments, parental pressure, or "everyone does it."

### Critical Reasons



**Primary Examples:**
Unawareness of citation rules and inability to incorporate materials.

### Core Solution

Comprehensive citation and formal approval from involved countarparts.



### Critical Drivers of Misconduct

Plagiarism often stems from poor time-management and lack of source-integration skills.



### The Golden Rule of Acknowledgment

Always acknowledge resources and include quotation marks for every direct quote used.

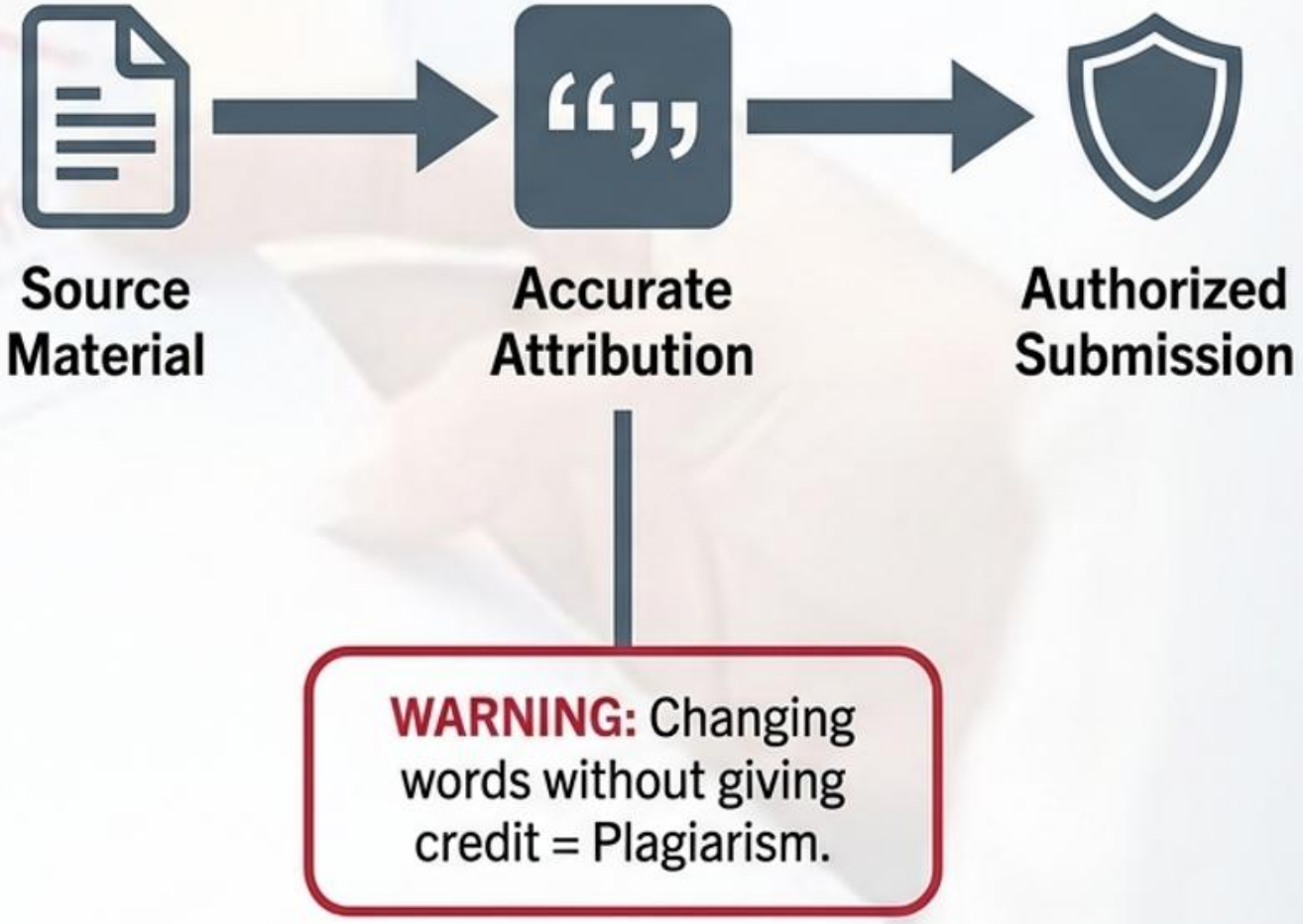


### Administrative Plagiarism

Submitting reports without approval from coordinators or counterparts is considered a form of plagiarism.

# Safeguarding Your Research Integrity

- **Acknowledge Everything:** Credit must be given for words, sentence structures, and underlying ideas.
- **Master Integration:** Learn the mechanical rules of incorporating sources seamlessly.
- **Zero Tolerance:** Forgetting to cite is indistinguishable from intentional theft.
- **Ultimate Takeaway:** Academic survival relies on precise attribution, not just original thought.



# AI Usage Policy in Research at Nile University



- How can students use AI effectively as a tool to help generate research ideas and approaches?
- How can students use AI effectively as a writing or editing tool for publications and their thesis?
- How accurate are the results from an AI source?
- How confidential is the process? Is a student giving away valuable ideas or research results to an open platform (like ChatGPT) before the topic is peer reviewed and published? Will you lose your intellectual property rights, such as patents?
- How can students improve their skills in using AI as a tool in research?
- Are students at risk of committing research misconduct in the form of plagiarism if they use a generative AI platform to write parts of their thesis?
- How can students (and other researchers) transparently indicate within their written works the contribution of generative AI or other assistive technologies?
- What restrictions do professional societies and publishers place on the use of AI in publications and in reviewing papers or proposals?

# Navigating AI in Research: Nile University's Policy Framework

Policy No. POL-RO-xx establishes the institutional framework for the ethical, transparent, and responsible use of AI.
AI is an assistive tool and must never replace human intellectual contribution or compromise academic integrity.

## CORE PRINCIPLES AND MANDATORY RULES




### AI as an Assistive Tool, Not an Author




AI systems cannot be listed as authors; all academic responsibility remains with humans.

### Mandatory Disclosure of AI Usage




Users must disclose the tool name, purpose, and extent of AI contribution in outputs.

### Human Oversight is Non-Negotiable

Content generation, analysis, or interpretation requires punctual human scrutiny and critical evaluation.




## OPERATIONAL STANDARDS AND PROHIBITIONS

### PERMITTED ASSISTIVE TASKS

AI may be used for language editing, summarization, coding assistance, and data visualization.




### STRICT PROHIBITIONS




Fully AI-generated work and AI-produced academic review reports are strictly prohibited.

### Data Privacy and Integrity

Confidential or sensitive data must not be shared with AI without approved protocols.




## STAKEHOLDER RESPONSIBILITIES

**Faculty & Researchers**

Verify all AI content and assume full responsibility for errors.

**Research Office**

Monitor compliance and provide guidance on ethical AI use.

**Ethics Committee**

Review the ethical implications of AI use in specific research projects.

# Sources/References


**Living guidelines on the responsible use of generative AI in research**

https://research-and-innovation.ec.europa.eu/document/download/2b6cf7e5-36ac-41cb-aab5-0d32050143dc_en?filename=ec_rtd_ai-guidelines.pdf

**Egypt's national guidelines for trustworthy and responsible AI**

https://ai.gov.eg/SynchedFiles/ar/Resources/Egypt%20National%20Guidelines%20for%20Trustworthy%20and%20Responsible%20AI.pdf

**Policy on the Responsible Use of Artificial Intelligence in Academic Texts**

https://www.sharjah.ac.ae/-/media/project/uos/sites/uos/graduate-studies/bylaws-and-regulations/policy-on-the-responsible-use-of-artificial-intelligence-in-academic-texts.pdf

**Ethics guide of generative AI use in the scientific research and publication process of higher education institutions**

https://proje.yok.gov.tr/documentFiles/17539645794.Y%C3%BCksek%C3%B6%C4%9Fretimde%20%C3%BCretken%20yapay%20zeka%20kullan%C4%B1m%C4%B1-en.pdf


Thank
you